SAHARA

Survival & Health Agent for Relief and Recovery Aid

India's First Multi-Agent AI Disaster Response System

3 AI Agents

Real-Time Data

15 sec Response

🏆 **AgentathonX 2026 — HiGen Labs | India's First Online AI Agent Hackathon**

THE PROBLEM

600M+

Indians affected by
climate disasters annually

70%

of rural India lacks
adequate healthcare

30 min

The critical window —
most deaths are preventable

When disaster strikes, three systems fail simultaneously:

1. Environmental monitoring is siloed — no one knows which health risks the disaster creates
2. Medical triage is disconnected — generic advice fails survivors facing flood-borne leptospirosis or heatstroke
3. Relief coordination is manual — survivors navigate 3 separate systems while terrified and alone

THE SOLUTION

SAHARA connects all three systems in one 15-second pipeline

Agent 1

Environment Scanner

- Live weather + AQI data
- Detects flood / heatwave / fire / storm
- Severity score 0–10
- Maps disaster ⟶ health risks

→

Agent 2

Health Triage

- Disaster-aware symptom analysis
- NOT generic — cross-refs disaster type
- Urgency: Low / Medium / High / EMERGENCY
- Immediate first-aid steps + red flags

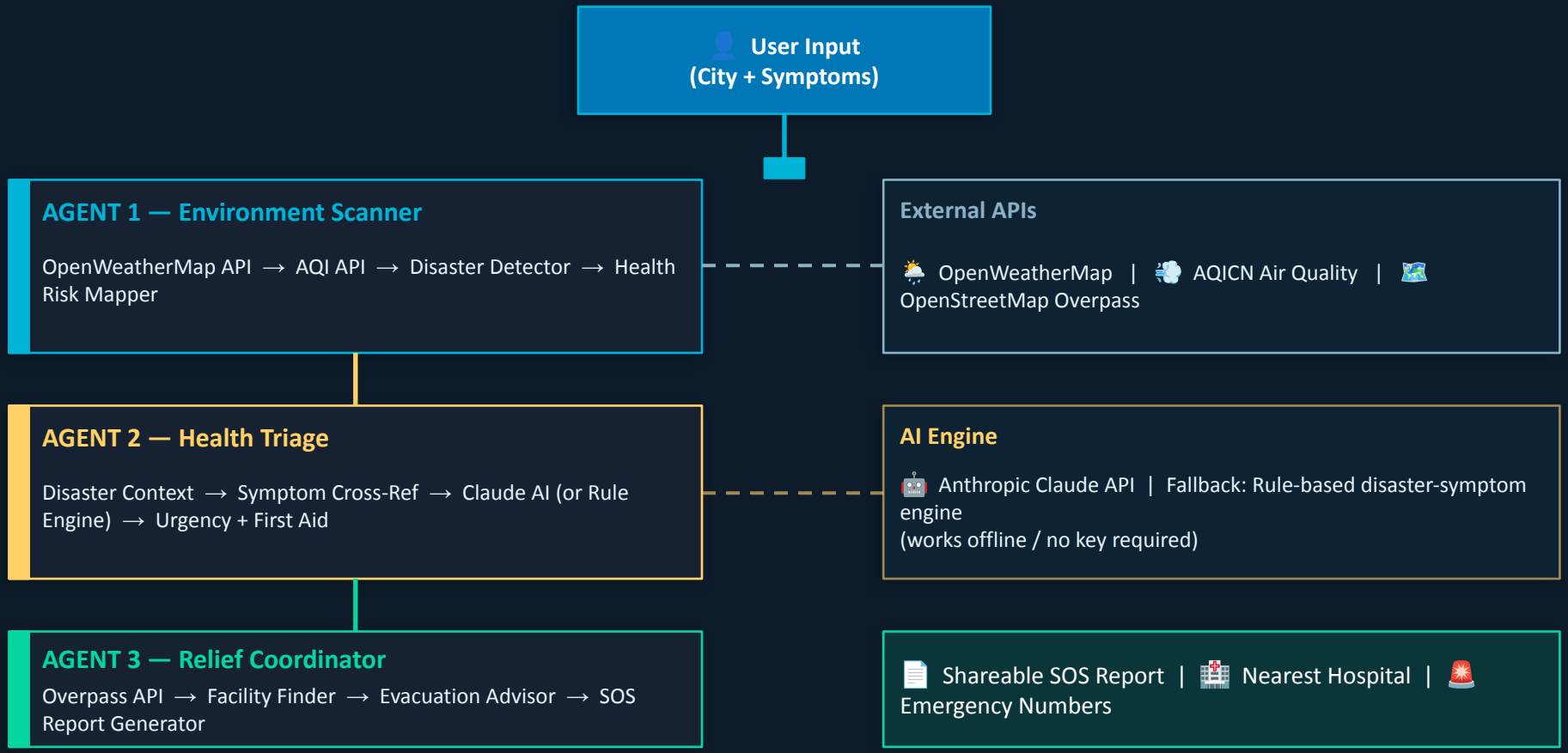
→

Agent 3

Relief Coordinator

- Nearest hospitals via OpenStreetMap
- Evacuation advice per disaster type
- Shareable SOS report (WhatsApp/SMS)
- Indian emergency numbers embedded

SYSTEM ARCHITECTURE & FLOWCHART



TECHNOLOGY STACK

Component	Technology	Purpose
Language	Python 3.10+	Core agent logic
UI Framework	Streamlit	Real-time web interface
AI Engine	Anthropic Claude API	claude-sonnet-4 for triage reasoning
Weather	OpenWeatherMap API	Live disaster detection
Air Quality	AQICN API	Real-time AQI data
Geospatial	OpenStreetMap Overpass API	Free, no key needed — hospitals & shelters
HTTP	Python requests	All API communication
Fallback AI	Custom Rule Engine	Disaster-symptom knowledge base
Cost	100% Free Tier APIs	Zero cost to run

MARKET OPPORTUNITY

$252B

Global Disaster Management Market

by 2030 (CAGR 6.8%)

$6.3B

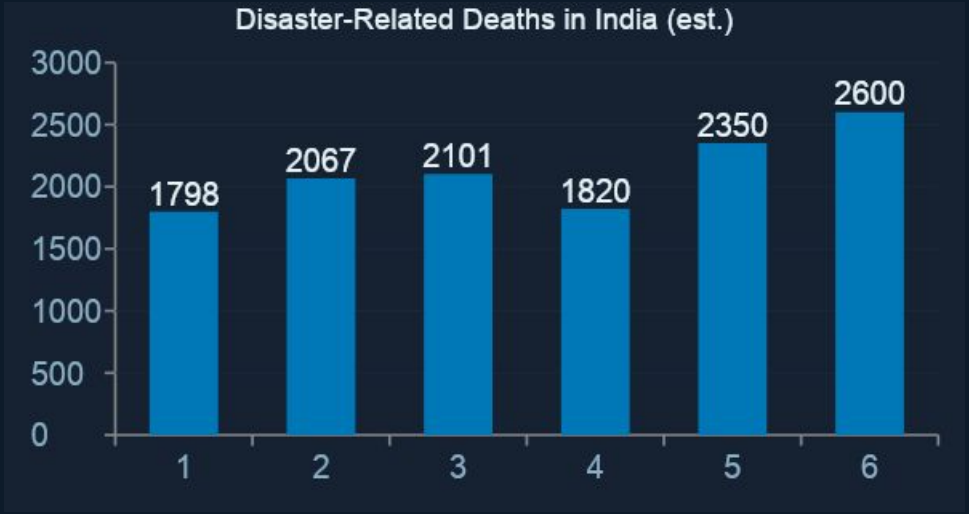
AI in Healthcare Market (India)

by 2027 (CAGR 40%)

1.1B+

Addressable Users (India)

smartphone + disaster-prone population



The Gap SAHARA Fills

NDRF / IMD	Alerts only — no health guidance
Generic symptom apps	No disaster context awareness
Hospital finders	No triage or environmental data
SAHARA ✅	All three — unified in 15 seconds

🚨 LIVE DEMO SCENARIO

📍 **A flood has just hit Silchar, Assam. A survivor opens SAHARA on their phone.**

1

Agent 1 — Environment Scanner

▸ *City: "Silchar"*

✅ Flood detected • Severity 8/10 • AQI: Unhealthy
Health risks: Leptospirosis, Cholera, Waterborne infections

2

Agent 2 — Health Triage

▸ *Survivor: "I have high fever and leg pain since yesterday"*

✅ Likely: Leptospirosis exposure (flood-specific)
Urgency: HIGH • Do NOT drink local water • Seek hospital within 24hrs

3

Agent 3 — Relief Coordinator

▸ *Coordinates from Agent 1 + Urgency from Agent 2*

✅ Nearest hospital: 2.3km (map link)
Evacuation route generated • SOS report ready to share

⚡ **Total pipeline time: ~15 seconds**

INNOVATION & REAL-WORLD IMPACT

What Makes SAHARA Different

Disaster-Aware Triage

Flood + fever = leptospirosis advice
— not generic 'drink water'

3-Agent Pipeline

Each agent enriches the next
with structured context

Graceful Degradation

Works without Anthropic key
Full fallback rule engine built-in

Zero Cost to Run

100% free APIs
Deployable anywhere in India

Lives It Can Save

Leptospirosis

50,000+ cases/yr in India — misdiagnosed as 'fever'

Heat Stroke

Kills 300+ every summer — most deaths preventable

Elderly & Rural

SAHARA works in any language, any device

No Doctor Needed

Offline fallback for zero-connectivity zones

CHALLENGES & LESSONS LEARNED

Challenges We Overcame

1

Generic vs. Contextual Advice

Early Agent 2 gave useless generic advice. Solution: build disaster-symptom cross-reference so advice is always tied to the specific disaster.

2

API Reliability During Disasters

Real disasters mean bad internet. Solution: complete fallback logic at every step — SAHARA never returns a blank screen.

3

3 APIs, 3 Different Formats

OpenWeather, AQICN, and Overpass return completely different structures. Solution: normalize all data into a single context object passed between agents.

Key Lessons

Agents Share Context, Not Just Results

SAHARA's power comes from Agent 2 receiving Agent 1's full environmental report — not just the disaster name.

Design for Failure First

A disaster tool that breaks in bad connectivity is worse than useless. Fallbacks are a core feature, not an afterthought.

Specificity Saves Lives

Generic advice is dangerous. 'Flood + fever = leptospirosis' is what makes SAHARA genuinely different from existing tools.

WHAT'S NEXT FOR SAHARA

v1.0 ✅ Today

- 3-Agent pipeline live
- OpenWeather + AQI + OSM
- Claude AI + rule-based fallback
- Streamlit web UI

v2.0 3 Months

- Voice input (Hindi/Tamil/Bengali)
- NDRF + IMD official API feeds
- Android app (offline-capable)
- SMS-based SOS without internet

v3.0 12 Months

- Predictive alerts 24–48hrs ahead
- Mesh network SOS relay
- State government integration
- 10 regional language support

SAHARA is not a prototype. It is a working, deployable system built in 24 hours that can save lives today.

Built for AgentathonX 2026